MiniStack Tutorial

Introduction

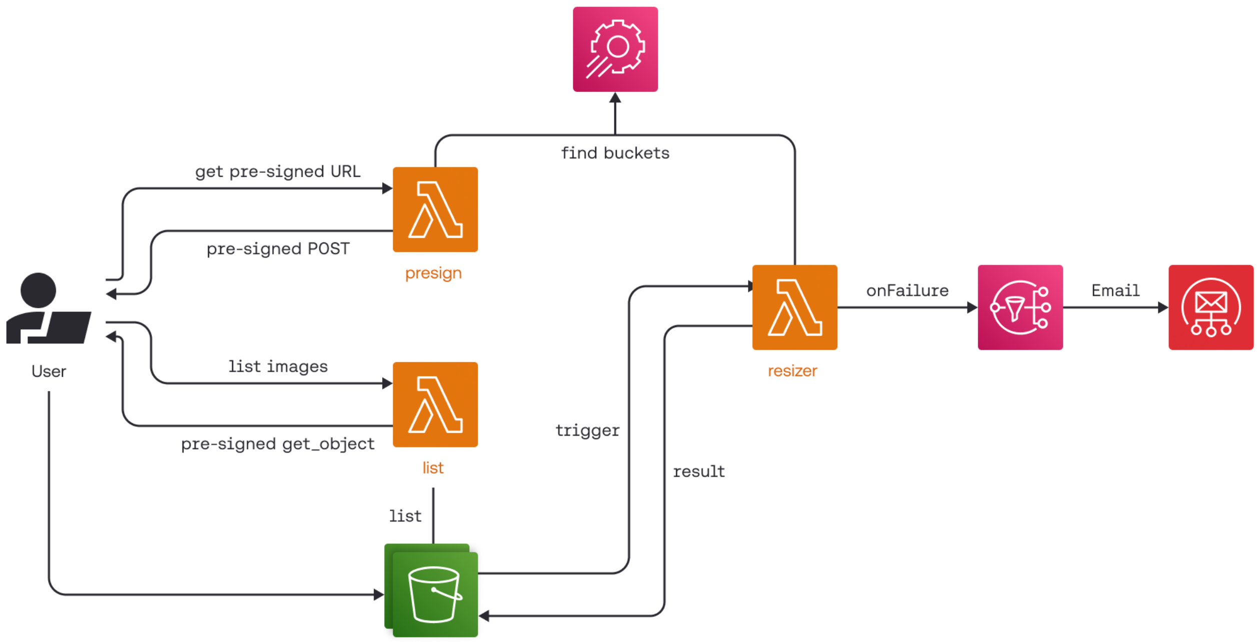
This is an app to resize images uploaded to S3 in a serverless way. A simple web fronted using HTML and JavaScript provides a way for users to upload images that are resized and listed. We use a Lambda to generate S3 pre-signed URLs so the upload form can upload directly to S3 rather than going through the Lambda. S3 bucket notifications are used to trigger a Python Lambda that runs image resizing. Another Lambda is used to list all uploaded and resized images, and provide pre-signed URLs for the browser to display them.

Here's a short summary of AWS service features we use:

- S3 bucket notifications to trigger a Lambda
- S3 pre-signed POST
- S3 website
- Lambda function URLs

> The tutorial original source is: https://github.com/localstack-samples/sample-serverless-image-resizer-s3-lambda

Architecture overview



Run On LBD Cluster

On the LBD cluster you do not need to create a virtual environment and you do not need to run `pip install`.

Start MiniStack in a shell:

```
ministack
```

Then in the project directory run:

```
bash ./run.sh
```

Open:

```
https://lbd.tuwien.ac.at/user/$USER/proxy/4566/webapp/index.html
```

Run Locally

As an alternative to the LBD cluster, you can run the project locally on your machine using MiniStack.

Setup:

```
python3 -m venv .env
source .env/bin/activate
pip install -r requirements.txt
```

Start MiniStack in a separate shell, then in the project directory run:

```
ministack
```

Then in the project directory run:

```
bash ./run.sh
```

Open:

```
http://localhost:4566/webapp/index.html
```

Notes

- `run.sh` creates the buckets, deploys the Lambdas, configures bucket notifications, and uploads the frontend.

- At the end of `run.sh`, the script prints the public web URL and the Lambda URLs it generated for the current environment.

Run integration tests

Once all resources are created on MiniStack, you can run the automated integration tests.

```
pytest tests/
```

Attribution

This project is adapted from LocalStack tutorial https://github.com/localstack-samples/sample-lambda-s3-image-resizer-hot-reload